Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

A. Radial artery

B. Ulnar artery

C. Princeps pollicis artery

D. Brachial artery

Reasoning + Answer:

<answer>A. Radial artery</answer>

<think>Reasoning process:

1. The anatomical snuffbox is a region in the wrist, bounded by the tendons of the extensor pollicis longus, extensor carpi radialis longus, and extensor carpi radialis brevis muscles.
2. The radial artery is located in the anatomical snuffbox, making it a likely candidate for injury in this region.
3. The radial artery is a major artery that supplies blood to the forearm and hand, and its injury could lead to significant bleeding.
4. The ulnar artery is located on the other side of the wrist, in the ulnar groove, and is less likely to be injured in this location.
5. The princeps pollicis artery is a small branch of the radial artery that supplies blood to the thumb, but it is not the primary artery responsible for the overall blood flow to the hand.
6. The brachial artery is a major artery that supplies blood to the upper arm, but it is not directly related to the anatomical snuffbox or the radial artery.

Considering these anatomical relationships, the radial artery is the most likely to be damaged in this scenario, leading to the patient's excessive bleeding.</think>